MILITARY SPECIFICATION

CLOTH, NETTING, PARACHUTE SKIRT EXTENSION, NYLON. RASCHEL KNIT

1. SCOPE

1.1 Scope.- This specification covers requirements for one type of raschel knitted nylon mesh net cloth.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics.

STANDARDS

FEDERAL

Fed. Std. No. 191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1491 - Glossary of Knitting Imperfections

PUBLICATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 3305

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3. REQUIREMENTS

3.1 Materials.-

3.1.1 Yarn.- The nylon yarn used in the manufacture of the netting shall be a bright, high tenacity, light and heat resistant polyamide prepared from hexamethylene and adipic acid or its derivatives. The nominal denier of yarn, prior to dyeing, shall be 840. The yarn shall have a melting point of 254° ± 10° Centigrade when tested as specified in 4.2.1. The yarn or the cloth netting shall not be bleached in any manner or process.

3.2 Color.- Unless otherwise specified (see 6.2) the color of the cloth netting shall be shade Olive Green 106.

3.2.1 Colorfastness.- The dyed and finished cloth netting shall show colorfastness equal to or better than the standard sample with respect to light, dry cleaning and laundering. In the event that a standard sample is not available the dyed and finished cloth netting shall show "good" fastness to these same properties when tested as specified in 4.2.4.

3.2.2 Matching.- The color of the dyed cloth netting shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight of 2800° Kelvin.

3.3 Physical requirements.-

3.3.1 Knit.- The netting shall be of the mesh type knitted on a single needle bar raschel machine (see 6.3). The cross section of the mesh leg (pillar) shall consist of 5 ends of 840 denier yarn (total nominal denier of 4200). The joining of the pillars (chain stitch) shall be accomplished by a single crossover of yarns during the knitting operation (see 6.3).

3.3.2 Mesh size.- The mesh size in the finished cloth netting shall be 3-3/4 inches by 3-3/4 inches (inside measurements) ± 1/4 inch when measured as specified in 4.2.2.2.

3.3.3 Width.- The overall width of the finished netting shall be 18 inches ± 1 inch when measured as specified in 4.2.2.2. In the preparation of this width the netting shall be cut so as to produce square meshes when the netting is hung in the lengthwise direction. One edge of this netting shall either have a knitted selvage or the edge shall be sewn (see 3.3.4) so that there shall be no loose ends along the full length (see 3.4) of that edge of the netting.

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3.3.4 Sewing of cut edge.- The loose ends of the cut edge shall be sewn, if necessary, along the full length of the netting (see 3.4) and the following sewing requirements shall apply:

(1) Thread - Single end of 840 denier nylon yarn for needle and loopers (see 3.1.1).

(2) Seam type - EFd-1.

(3) Stitch type - 504 - 505.

(4) Stitch gage - approximately 1/8 inch.

(5) Stitches per inch - 12 - 15.

3.4 Put-up.- Unless otherwise specified (see 6.2) the cloth netting shall be furnished on individual rolls in 500 feet - 5 + 25 feet lengths. There shall be no more than 10 joinings or mends (total) on each individual roll of netting.

3.5 Mending and joining.- Damages in the netting shall be mended by the patch method. The damaged portion of the netting shall be cut out and replaced with a piece of undamaged netting (see 3.3.2). For mending and joinings, a full leg (one side of square mesh) shall be overlapped with a full leg of the main piece of the cloth netting and secured with stitch type 304 (zig-zag) using the thread yarn specified in 3.3.4.

3.6 Fiber identification.- Each roll shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.7 Workmanship.- The finished cloth netting shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Certificate of compliance.- When certificates of compliance are submitted the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in the specification or applicable purchase documents. In addition, inspection shall be performed for the requirements listed in table I.

TABLE I.- Component tests

Characteristic	Paragraph reference	Test method
Material:		
Yarn	3.1.1	1/
Lusture	3.1.1	1/
Heat and light resistance	3.1.1	1/
Denier	3.1.1	1/
Melting point	3.1.1	1534 1/ 2/
Bleaching	3.1.1	1/

1/ Unless otherwise specified, a supplier's certificate of compliance shall be furnished and will be accepted for the requirements.

2/ Refers to test method of Fed. Std. No. 191.

4.2.2 Examination of the end item for visual and dimensional defects.- The defects found during these examinations shall be classified in accordance with 4.2.2.1 and 4.2.2.2.

4.2.2.1 Yard-by-yard examination.- The required yardage of each roll shall be visually inspected for defects as defined in MIL-STD-1491 and as listed herein. All defects shall be counted regardless of their proximity, one to another, except where two or more defects represent a single local condition of the netting, in which case only one defect shall be counted. A continuous defect shall be counted as one defect for each lengthwise three feet (1 yd) of fraction thereof in which it occurs. The sample unit for this examination shall be three feet (1 yd). The sample size shall be in accordance with inspection level II of MIL-STD-105. The AQL shall be 4.0 defects per 100 units (3 feet per yard). The lot size shall be expressed in units of 3 feet each (1 yd). An approximate equal number of feet shall be examined from each roll selected. The number of rolls over which the sample yardage shall be distributed shall be in accordance with Table II.

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Defects

Cut, break or tear.
Weak or thin area.
Abrasion mark.
Drop ply or drop stitch.
Thick or thin place - less than half or more than twice the normal thickness of the yarn.
Loose or tight knitting.
Mixed yarn.
Float, kink or miss - knit.
Selvage cut, torn or not sewn as required.
Stitch type (mending or joining) other than specified.
Spot or stain.
Meshes not within specified tolerance. 1/
Width not within specified tolerance. 2/

1/ Determination of mesh size. The mesh size shall be determined by hand measuring the lengths of the four sides of the pillars (chain stitches) that constitute the mesh. Slight tension shall be applied to the pillars (chain stitch) while measuring each length of the mesh size.

2/ Determination of overall width of netting. One continuous length of pillars (chain stitches) extending from the sewn or selvaged edge to the opposite edge of the completed net shall be measured by hand. Slight tension shall be applied while measuring the width of the net.

4.2.2.2 Overall examination.- Each defect listed below shall be counted not more than once in each roll examined. The sample unit for this examination shall be one roll. The sample size (number of rolls selected as sample) for this examination and the acceptance number shall be as shown in Table II. The rolls selected for the yard-by-yard examination shall be used for this examination.

Length of netting not within tolerances. 1/
Color not as specified.
Selvaged edge not knitted or sewn as required.
Selvage seam, stitch type and stitches per inch not as required.
More than 10 mends and joins (total) per roll of netting.
Off shade.
Overall uncleanness.

1/ Length of net. The length shall be measured along the sewn edge and the edge shall be straightened by slight tension prior to measuring or measurements shall be made over a measuring clock under normal winding tension.

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TABLE II

Lot Size in Yards (3 feet)	Sample Size in Rolls	Acceptance No.
Up to 1200 1/	3	0
1201 up to & including 3200	5	0
3201 up to & including 10,000	8	0
10,001 up to & including 35,000	13	0
35,001 up to & including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than 3 rolls, each roll shall be examined.

4.2.3 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging, packing and marking complies with the section 5 requirements in accordance with the provisions of PPP-P-1133.

4.2.4 Testing of the end item.- The methods of testing specified in Fed. Std. No. 191, wherever applicable and as listed in Table III shall be followed. The physical values specified in section 3 apply to the average of the results made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be one roll of netting. The lot size shall be expressed in units of one roll of netting each. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The sample size (number of sample units) shall be in accordance with the following:

Lot size (rolls of netting)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III.- Test Methods

Characteristic	Requirement paragraph	Test Method
Colorfastness to:		
Light	3.2.1	5660 1/
Drycleaning	3.2.1	5620 2/
Laundering	3.2.1	5614 3/
Knit	3.3.1	Visual 3/

1/ The time of exposure shall be 20 standard fading hours and the test specimen shall be 4 to 5 grams of the cloth net material.

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2/ Except that the test specimen shall be 4 to 5 grams of the cloth net material.

3/ Unless otherwise specified, a supplier's certificate of compliance shall be furnished and will be accepted for the requirements.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging.- Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C.- The cloth netting shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B or C.- The cloth, netting shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-1133.

6. NOTES

6.1 Intended use.- The cloth netting covered by this specification is intended for use as a parachute skirt extension.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Color if other than specified (see 3.2).
(c) Put-up if other than specified (see 3.4).
(d) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Knitting machine.- The following is for information only and is not a mandatory requirement (see 3.3.1). The prototype sample was knitted on a single needle bar, 5 guide bar Raschel machine, with approximately 12 needles per inch (24 gauge). The prototype sample had 5 mesh repeats in the completed net. The following chain notation may be used as a guide.

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BARS

1 (Front)	2 (Back)	3 (Back)	4 (Back)	5 (Back)
2-0	4-2	2-2	4-2	2-2
0-2	4-4 x 27	2-4 x 27	4-4 x 27	2-4 x 27
R	2-4	2-2	2-4	2-2
E	2-2	4-2	4-4	4-2
P	4-6	6-6	2-0	0-0
E	2-2	4-2	4-4	4-2
A	4-2	2-2	4-2	2-2
T	4-4 x 28	2-4 x 28	4-4 x 28	2-4 x 28
	2-4	4-4	2-4	2-2
	0-0	2-0	6-6	4-6
	2-4	4-4	2-4	2-2
	4-4	4-2	4-4	4-2

Custodian:

Army - GL

Preparing activity:

Army - GL

Project No. 8305-A209

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Cloth, Netting, Parachute Shirt Extension, Nylon, Raschel Knit MIL-C-43805(GL)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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